KRESLA PLATFORM DOCUMENTATION

Software Requirements Specification (SRS)

KRESLA

Making Real Estate Simple, Secure, and Accessible

Company: Kresla Technologies

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1. Executive Summary

Kresla is a modern digital real estate marketplace developed to connect landlords, tenants, buyers, property managers, and businesses on one secure platform.

The platform simplifies discovering, listing, renting, buying, and managing real estate through an intelligent marketplace powered by technology and location services.

Instead of relying on Facebook groups, classified advertisements, or word-of-mouth referrals, users gain access to a centralized and verified property ecosystem.

Kresla aims to become the trusted home of real estate transactions by providing transparency, convenience, security, and efficiency.

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2. Company Background

Kresla Technologies is a technology venture founded in **2023** with the objective of creating practical digital solutions for everyday challenges.

The company's first flagship product is **Kresla**, a comprehensive real estate marketplace designed to modernize property discovery and management.

The organization believes technology should remove unnecessary barriers while creating trust between service providers and customers.

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3. Vision

To become Africa's most trusted digital real estate marketplace that connects people with properties through innovation, transparency, and technology.

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4. Mission

To simplify the process of finding, listing, renting, buying, and managing properties while creating a secure and trustworthy ecosystem for landlords, tenants, businesses, and property investors.

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5. Problem Statement

Finding suitable accommodation remains difficult for many people.

Current methods include:

- Facebook Marketplace
- WhatsApp groups
- Classified advertisements
- Estate agents
- Friends and family referrals
- Physical property searches

These methods often present challenges such as:

- Fake property listings
- Fraud and scams
- Outdated advertisements
- Poor communication
- Lack of verified landlords
- Limited search options
- Time-consuming property hunting
- Inaccurate location information

Landlords also face difficulties advertising properties to the right audience while managing inquiries efficiently.

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6. Proposed Solution

Kresla provides one digital platform where landlords can advertise properties and tenants can search for verified listings using intelligent search and filtering tools.

The platform offers:

- Verified user accounts
- Property listings with images
- Interactive maps
- Property ratings
- Real-time messaging
- Saved favorites
- Secure administration
- Location-based searches
- 3D virtual property viewing

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7. Objectives

The objectives of Kresla include:

- Simplify property searching
- Improve landlord visibility
- Reduce fraud
- Improve communication
- Digitize rental management
- Create trusted property listings
- Enable smart property discovery
- Improve transparency in the real estate market

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8. Scope of the Platform

The platform supports:

- Residential rentals
- Commercial rentals
- Property sales
- Vacation accommodation
- Lodges
- Student accommodation
- Offices
- Shops
- Warehouses
- Industrial properties
- Hotels
- Guest houses
- Apartments
- Stands
- Shared accommodation

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9. Target Users

Tenants

Individuals searching for accommodation.

Landlords

Property owners listing available properties.

Property Managers

Organizations managing multiple rental units.

Real Estate Agencies

Companies marketing properties.

Buyers

Individuals purchasing properties.

Sellers

Individuals selling property.

Businesses

Companies seeking commercial premises.

Platform Administrators

Personnel responsible for platform management.

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10. Functional Requirements

Authentication

Users should be able to:

- Explore the platform before registration
- Register
- Sign in
- Sign out
- Reset password
- Verify email
- Manage account

User Profile

Each profile should contain:

- Profile picture

- Full name
- Contact information
- Bio (optional)
- Account type
- Ratings
- Verification badge
- Posted properties
- Saved properties

Property Listings

Each property listing should include:

- Property title
- Property description
- Rental or sale price
- Number of bedrooms
- Number of bathrooms
- Parking availability
- Property size
- Property type
- Images
- Videos
- 3D virtual tour
- GPS coordinates
- Interactive map
- Availability status
- Date posted
- Landlord profile

Search and Filtering

Users should filter properties by:

- Price
- Bedrooms
- Bathrooms
- Property type
- Suburb
- City
- Province
- Country
- Stand size
- Property size

- Furnished or unfurnished
- Parking availability
- Pet friendly
- Property availability
- Listing date

Location-Based Search

The platform should provide:

- Interactive maps
- GPS navigation
- Nearby schools
- Nearby hospitals
- Nearby shopping centers
- Nearby transport routes
- Distance calculations

Property Gallery

Properties should support:

- High-resolution photos
- Image sliders
- Full-screen gallery
- Video walkthroughs
- 360° images
- 3D house tours

Messaging System

The platform should provide secure messaging between landlords and tenants.

Features include:

- Real-time chat
- Image sharing
- File attachments
- Read receipts
- Conversation history
- Contact requests
- Notifications

Favorites

Users should be able to:

- Save favorite properties
- Remove favorites
- Compare saved properties
- Receive updates when prices change

Property Ratings

Users can:

- Upvote properties
- Downvote properties
- Leave ratings
- Leave reviews
- Report inaccurate information

Ratings improve transparency and help future tenants make informed decisions.

Property Reporting

Users should report:

- Fraud
- Duplicate listings
- False information
- Offensive content
- Sold properties still listed
- Spam

Landlord Property Management

Landlords should be able to:

- Add properties
- Edit properties
- Delete properties
- Mark properties as rented
- Mark properties as sold

- Pause listings
- Renew listings
- View property statistics
- View inquiries

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11. Non-Functional Requirements

The system should provide:

- High availability
- Fast performance
- Secure authentication
- Responsive design
- Cloud storage
- Mobile compatibility
- Data encryption
- Automatic backups
- Scalable architecture
- Reliable uptime

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12. System Modules

Authentication Module

Handles user registration and login.

User Management Module

Manages user accounts.

Property Management Module

Handles property listings.

Search Module

Provides intelligent filtering.

Maps Module

Provides location services.

Messaging Module

Supports landlord-tenant communication.

Ratings Module

Handles reviews and votes.

Notifications Module

Sends alerts.

Reports Module

Handles abuse reports.

Administration Module

Provides platform management tools.

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13. User Roles

Tenant

Can:

- Search properties
- Save favorites
- Contact landlords
- Rate listings
- Report fraud

Landlord

Can:

- List properties
- Edit listings
- Remove listings
- Receive inquiries
- Chat with tenants
- View analytics

Administrator

Can:

- Manage users
- Remove fake listings
- Moderate reports
- View analytics
- Verify landlords
- Suspend accounts
- Export reports
- Monitor platform activity

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14. Platform Features

- Secure Login
- User Profiles
- Property Listings
- Image Galleries
- 3D Virtual Tours
- Interactive Maps
- GPS Integration
- Property Ratings
- Reviews
- Saved Properties
- Messaging
- Push Notifications
- Email Notifications
- Search Filters
- Advanced Search
- Fraud Reporting
- Verified Users
- Listing Management
- Dashboard Analytics
- Activity Logs

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15. Property Categories

Kresla supports:

- Houses
- Apartments
- Flats

- Rooms
- Student Accommodation
- Lodges
- Hotels
- Guest Houses
- Commercial Buildings
- Shops
- Offices
- Warehouses
- Industrial Buildings
- Shopping Spaces
- Land
- Residential Stands
- Agricultural Land
- Vacation Homes

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16. User Journey

Tenant Journey

1. Open Kresla
2. Explore available listings
3. Register or sign in
4. Complete profile
5. Search properties
6. Apply filters
7. View property details
8. View gallery and 3D tour
9. Contact landlord
10. Save favorite property
11. Arrange viewing
12. Rent or purchase property

Landlord Journey

1. Register
2. Verify account
3. Complete profile
4. Create property listing
5. Upload photos and videos
6. Add map location
7. Publish listing
8. Receive inquiries
9. Chat with tenants

10. Mark property as rented or sold

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17. Administrator Dashboard

Administrators should have access to:

User Analytics

- Total users
- Active users
- New registrations
- Verified accounts

Property Analytics

- Total listings
- Active listings
- Sold properties
- Rented properties
- Pending approvals

Reports

- Fraud reports
- Abuse reports
- Duplicate listings
- User complaints

Revenue Dashboard (if monetized)

- Subscription revenue
- Premium listing revenue
- Advertisement revenue
- Transaction statistics

System Monitoring

- Login activity
- User activity logs
- Property upload trends
- Geographic distribution of listings
- Search trends
- Most viewed properties
- Most saved properties

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18. Security Requirements

The platform should include:

- Secure authentication
- Password encryption
- Multi-factor authentication (optional)
- Email verification
- Phone verification
- Secure cloud storage
- Activity logs
- Role-based permissions
- Secure messaging
- Data protection compliance

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19. Future Enhancements

Future releases may include:

- AI-powered property recommendations
- Mortgage calculators
- Digital lease agreements
- Online rent payments
- Utility bill management
- Property valuation tools
- AI chatbot support
- Property demand forecasting
- Document verification
- Virtual reality (VR) property tours
- Real estate investment analytics

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20. Technology Recommendations

Frontend

- React
- Next.js
- Flutter (Mobile)

Backend

- Supabase

- Node.js
- PostgreSQL

Maps

- Google Maps API
- OpenStreetMap

Storage

- Supabase Storage

Authentication

- Supabase Auth

Notifications

- Firebase Cloud Messaging

Hosting

- Vercel
- Supabase Cloud

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21. Expected Benefits

For Tenants

- Faster property discovery
- Verified listings
- Better communication
- Reduced fraud
- Improved decision-making

For Landlords

- Wider audience reach
- Easier property management
- Increased occupancy
- Faster tenant acquisition

For Businesses

- Better visibility
- Easier commercial property discovery
- Improved customer engagement

For Administrators

- Centralized management
- Real-time analytics
- Fraud detection
- Better platform governance

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22. Conclusion

Kresla represents a next-generation digital real estate marketplace built to transform the way people discover, list, rent, buy, and manage properties. By combining intelligent search, location-based services, verified users, secure communication, rich media, and powerful administrative tools, Kresla creates a transparent and trusted ecosystem for every participant in the property market.

Designed with scalability and innovation in mind, the platform lays the foundation for future capabilities such as AI-powered recommendations, digital contracts, integrated payments, and immersive virtual property experiences. Through technology, Kresla aims to make every property search simpler, every listing more credible, and every real estate transaction more efficient.